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Detection system and detection method

Abstract

A detection system and a detection method are provided. The detection method includes: obtaining first, second and third keypoints of a face by a keypoint acquisition module based on an image containing the face, the keypoint acquisition module obtaining the third keypoint based on a predetermined position on a midline of a human face and the first and the second keypoints based on two paired positions outside the midline; obtaining a vector by a first calculation module based on the first and second keypoints; obtaining a two-variable linear function by a second calculation module based on the vector and the third keypoint; and substituting, by a determination module, the coordinates of the first keypoint and the coordinates of the second keypoint into the two-variable linear function to obtain first and second values, respectively, and determining the state of the face based on the first and second values.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This non-provisional application claims priority under 35 U.S.C. § 119(a) to Patent Application No. 112102471 filed in Taiwan, R.O.C. on Jan. 18, 2023, the entire contents of which are hereby incorporated by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to the field of image identification, and particularly relates to a technology for determining a state of a human face in an image by applying keypoints on the human face in the image.

Related Art

(3) Traditionally, the pupil distance of human eyes in a human face is a common index for front face evaluation, but this index does not take into account the distance problem in human face detection. The pupil distance of the distant front face may be the same as that of the close side face, so it is still unable to correctly select a front face sample for human face identification.

SUMMARY

(4) In view of this, some embodiments of the present disclosure provide a training system, a training method and an identification system, aiming at solving existing technical problems.

(5) Some embodiments of the present disclosure provide a detection system. The detection system includes a keypoint acquisition module, a first calculation module, a second calculation module and a determination module; the keypoint acquisition module is configured to receive an image containing a face and obtain a first keypoint, a second keypoint and a third keypoint of the above face based on the above image, where the keypoint acquisition module obtains the third keypoint based on a predetermined position on a midline of a human face and the first keypoint and the second keypoint based on two paired positions outside the midline of the human face; the first calculation module is configured to obtain a vector based on the first keypoint and the second keypoint; the second calculation module is configured to obtain a two-variable linear function based on the above vector and the third keypoint; and the determination module is configured to substitute coordinates of the first keypoint and coordinates of the second keypoint into the above two-variable linear function to obtain first and second values, respectively, and determine the state

of the face based on the first and second values.

(6) Some embodiments of the present disclosure provide a detection method applicable to the detection system including a keypoint acquisition module, a first calculation module, a second calculation module and a determination module; the detection method includes the following steps: receiving an image containing a face by the keypoint acquisition module, and obtaining a first keypoint, a second keypoint and a third keypoint of the face based on the above image, where the keypoint acquisition module obtains the third keypoint based on a predetermined position on a midline of a human face and the first keypoint and the second keypoint based on two paired positions outside the midline of the human face; obtaining a vector by the first calculation module based on the first keypoint and the second keypoint; obtaining a two-variable linear function by the second calculation module based on the above vector and the third keypoint; and substituting, by the determination module, the coordinates of the first keypoint and the coordinates of the second keypoint into the two-variable linear function to obtain first and second values, respectively, and determining the state of the face based on the first and second values.

(7) Based on the above detection system and detection method provided by some embodiments of the present disclosure, the state of the face in the image can be quickly determined by performing simple algebraic operation and determination on the detected first keypoint, second keypoint and third keypoint of the face.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a detection system block diagram drawn according to some embodiments of the present disclosure.

(2) FIG. 2 is a human face and keypoint schematic diagram drawn according to some embodiments of the present disclosure.

(3) FIG. 3 is a human face and keypoint schematic diagram drawn according to some embodiments of the present disclosure.

(4) FIG. 4-1 is a predetermined position and human face corresponding schematic diagram drawn according to some embodiments of the present disclosure.

(5) FIG. 4-2 is a predetermined position and human face corresponding schematic diagram drawn according to some embodiments of the present disclosure.

(6) FIG. 5 is a keypoint acquisition module block diagram drawn according to some embodiments of the present disclosure.

(7) FIG. 6 is a structure schematic diagram of an electronic device drawn according to some embodiments of the present disclosure.

(8) FIG. 7 is a detection method flowchart drawn according to some embodiments of the present disclosure.

(9) FIG. 8 is a detection method flowchart drawn according to some embodiments of the present disclosure.

(10) FIG. 9 is a detection method flowchart drawn according to some embodiments of the present disclosure.

(11) FIG. 10 is a detection method flowchart drawn according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(12) The above and other technical contents, features and functions of the present disclosure will be clearly presented in the following detailed description of the embodiments in conjunction with the accompanying drawings. The thickness or size of each element in the figures is expressed in an exaggerated, omitted or approximate way for those familiar with the art to understand and read, and

the size of each element is not entirely its actual size, and is not used to limit the implementation of the present disclosure, so it has no technical significance, and any modification of structure, change of proportion or adjustment of size, without affecting the efficacy and purpose of the present disclosure, it shall still fall within the scope of the technical content disclosed in the present disclosure. The same reference numerals in all diagrams will be used to represent the same or similar elements. The term “connection” mentioned in the following embodiments can refer to any direct or indirect, wired or wireless connection means. In this article, the “first” or “second” and other similar ordinal words described are used to distinguish or refer to the same or similar elements or structures, and do not necessarily imply the order of these elements in the system. It is to be understood that in some cases or configurations, ordinal words can be used interchangeably without affecting the implementation of the present disclosure.

(13) FIG. 1 is a detection system block diagram drawn according to some embodiments of the present disclosure. FIG. 2 is a human face and keypoint schematic diagram drawn according to some embodiments of the present disclosure. FIG. 3 is a human face and keypoint schematic diagram drawn according to some embodiments of the present disclosure. Referring to FIG. 1-FIG. 3 together, a detection system **100** includes a keypoint acquisition module **101**, a first calculation module **102**, a second calculation module **103** and a determination module **104**.

(14) The keypoint acquisition module **101** is configured to receive an image **105** containing a face, and the image **105** is an image **201** or an image **301**. The keypoint acquisition module **101** is configured to obtain a first keypoint, a second keypoint and a third keypoint of the face based on a plurality of predetermined positions and the above received image **105** containing the face. The keypoint acquisition module **101** obtains the third keypoint based on the predetermined position on a midline of a human face and the first keypoint and the second keypoint based on two predetermined positions paired outside the midline of the human face.

(15) In some embodiments of the present disclosure, the above plurality of predetermined positions include positions of the nose and two eyes, the position of the nose is on the midline of the human face, and the positions of the two eyes are paired outside the midline of the human face. As shown in the embodiment drawn in FIG. 2, the keypoint acquisition module **101** obtains a first keypoint **202** and a second keypoint **203** of the image **201** corresponding to the positions of the two eyes and a third keypoint **204** corresponding to the position of the nose based on the predetermined positions of the nose and the two eyes. The aforementioned action of obtaining the first keypoint **202** and the second keypoint **203** of the image **201** corresponding to the positions of the two eyes and the third keypoint **204** corresponding to the position of the nose indicates obtaining the coordinates of each of the first keypoint **202**, the second keypoint **203** and the third keypoint **204**. The third keypoint **204** corresponding to the position of the nose is on a midline **206** of the human face. As shown in the embodiment drawn in FIG. 3, the above plurality of predetermined positions include positions of the nose and the two eyes. The keypoint acquisition module **101** obtains a first keypoint **302** and a second keypoint **303** of the image **301** corresponding to the positions of the two eyes and a third keypoint **304** corresponding to the position of the nose based on the positions of the nose and the eyes. The aforementioned action of obtaining the first keypoint **302** and the second keypoint **303** of the image **301** corresponding to the positions of the two eyes and the third keypoint **304** corresponding to the position of the nose indicates obtaining the coordinates of each of the first keypoint **302**, the second keypoint **303** and the third keypoint **304**. The third keypoint **304** corresponding to the position of the nose is on a midline **306** of the human face.

(16) In some embodiments of the present disclosure, the above plurality of predetermined positions include the positions of the nose, the left corner of the mouth and the right corner of the mouth, the position of the nose is on the midline of the human face, and the positions of the left corner of the mouth and the right corner of the mouth are paired outside the midline of the human face. As shown in the embodiment drawn in FIG. 2, the keypoint acquisition module **101** obtains a first keypoint **207** and a second keypoint **208** of the image **201** corresponding to the left corner of the

mouth and the right corner of the mouth, and a third keypoint **204** corresponding to the position of the nose based on the predetermined positions of the nose, the left corner of the mouth and the right corner of the mouth. As shown in the embodiment drawn in FIG. 3, the above plurality of predetermined positions include the positions of the nose, the left corner of the mouth and the right corner of the mouth. The keypoint acquisition module **101** obtains a first keypoint **307** and a second keypoint **308** of the image **301** corresponding to the left corner of the mouth and the right corner of the mouth, and a third keypoint **304** corresponding to the position of the nose based on the positions of the nose, the left corner of the mouth and the right corner of the mouth.

(17) FIG. 5 is a keypoint acquisition module block diagram drawn according to some embodiments of the present disclosure. Referring to FIG. 5, in the embodiment drawn in FIG. 5, the keypoint acquisition module **101** includes a trained neural network module **1011**. The trained neural network module **1011** is configured to receive the above image **105** containing the face and output the coordinates of each of the first keypoint, the second keypoint and the third keypoint of the face so as to enable the keypoint acquisition module **101** to obtain the first keypoint, the second keypoint and the third keypoint.

(18) In some embodiments of the present disclosure, the above neural network module **1011** includes a trained multi-task cascaded convolutional networks, and the trained multi-task cascaded convolutional networks can output keypoint coordinates corresponding to the positions of the nose, the left corner of the mouth, the right corner of the mouth and the two eyes on the image **105** of the face after receiving the image **105** containing the face. Then the keypoint acquisition module **101** obtains the first keypoint, the second keypoint and the third keypoint from the output of the multi-task cascaded convolutional networks based on the plurality of predetermined positions.

(19) FIG. 4-1 is a predetermined position and human face corresponding schematic diagram drawn according to some embodiments of the present disclosure. FIG. 4-2 is a predetermined position and human face corresponding schematic diagram drawn according to some embodiments of the present disclosure. In some embodiments of the present disclosure, the above plurality of predetermined positions are selected from a predetermined position **4-1** to a predetermined position **4-68**. As shown in FIG. 4-2, a predetermined position **4-9**, a predetermined position **4-28** to a predetermined position **4-31**, a predetermined position **4-34**, a predetermined position **4-52**, a predetermined position **4-58**, a predetermined position **4-63** and a predetermined position **4-67** are on the midline of the human face. The predetermined position **4-1** to a predetermined position **4-8**, a predetermined position **4-10** to a predetermined position **4-27**, a predetermined position **4-32** to a predetermined position **4-33**, a predetermined positions **4-35** to a predetermined position **4-51**, a predetermined position **4-53** to a predetermined position **4-57**, a predetermined position **4-59** to a predetermined position **4-62**, a predetermined position **4-64** to a predetermined position **4-66** and a predetermined position **4-68** drawn in FIG. 4-1 and FIG. 4-2, are paired at symmetrical positions on the human face. In some embodiments of the present disclosure, the third keypoint is selected from one of the above predetermined position **4-9**, predetermined position **4-28** to predetermined position **4-31**, predetermined position **4-34**, predetermined position **4-52**, predetermined position **4-58**, predetermined position **4-63** and predetermined position **4-67**, e.g. the predetermined position **4-28**. The first keypoint and the second keypoint are selected from the paired predetermined positions in the predetermined position **4-1** to predetermined position **4-8**, the predetermined position **4-10** to predetermined position **4-27**, the predetermined position **4-32** to predetermined position **4-33**, the predetermined position **4-35** to predetermined position **4-51**, the predetermined position **4-53** to predetermined position **4-57**, the predetermined position **4-59** to predetermined position **4-62**, the predetermined position **4-64** to predetermined position **4-66** and the predetermined position **4-68**, e.g. a predetermined position **4-3** and a predetermined position **4-15**.

(20) In some embodiments of the present disclosure, the keypoint acquisition module **101** uses the FacemarkLBF class in OpenCV to obtain keypoint coordinates on the human face of the image **105** corresponding to the above predetermined positions **4-1** to predetermined positions **4-68**, and then

the keypoint acquisition module **101** selects coordinates corresponding to the above predetermined positions from the obtained keypoint coordinates corresponding to the above predetermined positions **4-1** to predetermined positions **4-68** based on the plurality of predetermined positions so as to obtain the first keypoint, the second keypoint and the third keypoint. It is to be noted that the keypoint acquisition module **101** also uses modules of other image processing software to obtain the keypoint coordinates on the face of the image **105** corresponding to the above predetermined positions **4-1** to predetermined positions **4-68**. The present disclosure is not limited by the above FacemarkLBF class in OpenCV.

(21) The detection method according to the embodiments of the present invention and how modules of the detection system **100** cooperate to operate are described below in detail with reference to the drawings.

(22) FIG. 7 is a detection method flowchart drawn according to some embodiments of the present disclosure. Referring to FIGS. 1-3 and FIG. 7 together, in the embodiment drawn in FIG. 7, the detection method includes steps S701-S704. In step S701, the image **105** containing the face is received by the keypoint acquisition module **101**. Then, the first keypoint, the second keypoint and the third keypoint of the above face are obtained based on the image **105** by the keypoint acquisition module **101**, the third keypoint (e.g. the third keypoint **204** or the third keypoint **304**) is obtained by the keypoint acquisition module **101** based on the predetermined position (e.g. the position of the nose drawn in FIG. 2) on the midline of the human face, and the first keypoint (e.g. the first keypoint **202** or the first keypoint **302**) and the second keypoint (e.g. the second keypoint **203** corresponding to the first keypoint **202** or the second keypoint **303** corresponding to the first keypoint **302**) are obtained based on two positions (e.g. positions of two eyes drawn in FIG. 2) paired outside the midline of the human face.

(23) In the step S702, a vector is obtained by the first calculation module **102** based on the first keypoint and the second keypoint. For example, a vector **205** is obtained by the first calculation module **102** based on the first keypoint **202** and the second keypoint **203**. For example, a vector **209** is obtained by the first calculation module **102** based on the first keypoint **207** and the second keypoint **208**. For example, a vector **305** is obtained by the first calculation module **102** based on the first keypoint **302** and the second keypoint **303**. For example, a vector **309** is obtained by the first calculation module **102** based on the first keypoint **307** and the second keypoint **308**.

(24) In the step S703, the two-variable linear function is obtained by the second calculation module **103** based on the above vector and the third keypoint. In the step S704, the coordinates of the first keypoint and the coordinates of the second keypoint are substituted into the above two-variable linear function by the determination module **104** to obtain first and second values respectively. The state of the face in the image **105** is determined by the determination module **104** based on the above first and second values. In some embodiments of the present disclosure, the state of the above face includes a front face state and a side face state.

(25) FIG. 8 is a detection method flowchart drawn according to some embodiments of the present disclosure. Referring to FIG. 1, FIG. 7 and FIG. 8 together, in the embodiment drawn in FIG. 8, the above step S704 includes steps S801 to S803. In the step S801, whether the product of the first and second values is a positive number is determined by the determination module **104**. If the product of the first and second values is a positive number, the step S802 is carried out, and if the product of the first and second value is a negative number, the step S803 is carried out. In the step S802, the face is determined as the side face state by the determination module **104** responding to the product of the first and second values as a positive number. In the step S803, the face is determined as the front face state by the determination module **104** responding to the product of the first and second values as a negative number.

(26) The first keypoint **202** and the second keypoint **203** corresponding to the positions of two eyes in the image **201** and the third keypoint **204** corresponding to the position of the nose in FIG. 2, and the first keypoint **302** and the second keypoint **303** corresponding to the positions of the two eyes

in the image **301** and the third keypoint **304** corresponding to the position of the nose in FIG. 3 are treated as examples for illustrating, the coordinates of the first keypoint **202** are (54,56), the coordinates of the second keypoint **203** are (74,56), the coordinates of the third keypoint **204** are (64,64), the coordinates of the first keypoint **302** are (73,51), the coordinates of the second keypoint **303** are (88,55), and the coordinates of the third keypoint **304** are (91,68); the above coordinates are pixel coordinates of the image **105**, namely the coordinates of the vertex of an upper left corner of the image **105** are (0,0); a first coordinate component of the coordinates refers to a position counted from left to right by taking the vertex of the upper left corner of the image **105** as a reference; and a second coordinate component of the coordinates refers to a position counted from top to bottom by taking the vertex of the upper left corner of the image **105** as a reference. It is to be noted that since the method of this invention does not produce different results due to the choice of different coordinate systems, so that the present disclosure is not limited by the above pixel coordinates.

(27) FIG. 9 is a detection method flowchart drawn according to some embodiments of the present disclosure. Referring to FIGS. 1-3 and FIG. 9 together, in the embodiment drawn in FIG. 9, the above step **S702** includes a step **S901**. In the step **S901**, the coordinates of the first keypoint are subtracted from the second keypoint to obtain the above vector by the first calculation module **102**. By taking the above examples for illustrating as an example, the coordinates (54,56) of the first keypoint **202** are subtracted from the coordinates (74,56) of the second keypoint **203** to obtain a vector **205** (20,0) by the first calculation module **102**. The coordinates (73,51) of the first keypoint **302** are subtracted from coordinates (88,55) of the second keypoint **303** to obtain a vector **305** (15,4) by the first calculation module **102**.

(28) FIG. 10 is a detection method flowchart drawn according to some embodiments of the present disclosure. Referring to FIGS. 1-3, FIGS. 1-8 and FIG. 10 together, in the embodiment drawn in FIG. 10, the above step **S703** includes steps **S1001-S1003**. In the step **S1001**, the first component of the above vector is set as a coefficient of a first variable of an intermediate two-variable linear equation by the second calculation module **103**, and a second component of the above vector is set as a coefficient of a second variable of the above intermediate two-variable linear equation.

(29) In the step **S1002**, the first coordinates and the second coordinates of the third keypoint are substituted into the above intermediate two-variable linear equation to obtain a constant term of the intermediate two-variable linear equation by the second calculation module **103**. In the step **S1003**, the first component of the above vector is set as a coefficient of the first variable of the two-variable linear function, the second component of the above vector is set as a coefficient of the second variable of the above two-variable linear function, and the constant term of the above intermediate two-variable linear equation is set as a constant term of the above two-variable linear function.

(30) The first keypoint **202** (the coordinates are (54,56)) and the second keypoint **203** (the coordinates are (74,56)) corresponding to the positions of two eyes in the above image **201** in FIG. 2, the third keypoint **204** (the coordinates are (64,64)) corresponding to the position of the nose and the vector **205** (the value is (20,0)) are illustrated as follows.

(31) In the step **1001**, the first component of the vector **205** is substituted into the coefficient $\alpha_{\text{sub.1}}$ of the first variable x of the following intermediate two-variable linear equation (1) by the second calculation module **103**, the second component 0 of the vector **205** is substituted into the coefficient $\beta_{\text{sub.1}}$ of the second variable y of the following intermediate two-variable linear equation (1),

(32) ##STR00001## so as to obtain an equation $20x + y_{\text{sub.1}} = 0$. In the step **1002**, a first coordinate component 64 and a second coordinate component 64 of the third keypoint **204** are substituted into the first variable x and the second variable y of the above intermediate two-variable linear equation by the second calculation module **103**, so as to obtain a value of the constant term $\gamma_{\text{sub.1}}$ of the intermediate two-variable linear equation to be -1280. In the step **S1003**, the first component of the

vector **205** is substituted into the coefficient $\alpha_{\text{sub.2}}$ of the first variable x of the following two-variable linear function (2), the second component of the vector **205** is substituted into the coefficient $\beta_{\text{sub.2}}$ of the second variable y of the following two-variable linear function (2), and the constant term $\gamma_{\text{sub.1}} = -1280$ of the above intermediate two-variable linear function is substituted into the constant term $\gamma_{\text{sub.2}}$ in the following two-variable linear function (2), (33) ##STR00002## so as to obtain a two-variable linear function $20x - 1280$.

(34) It is to be noted that the coordinates (54,56) of the first keypoint **202** are substituted into the first variable x and the second variable y of the two-variable linear function $20x - 1280$ respectively to obtain a first value of -200 by the determination module **104**, and the coordinates (74,56) of the second keypoint **203** are substituted into the first variable x and the second variable y of the two-variable linear function $20x - 1280$ to obtain a second value of 200 . The face in the image **201** is determined as the front face state by the determination module **104** responding to the product of the first value -200 and the second value 200 as a negative number.

(35) The first keypoint **302** (the coordinates are (73,51)) and the second keypoint **303** (the coordinates are (88,55)) corresponding to the positions of two eyes in the above image **301** in FIG. 3, the third keypoint **304** (the coordinates are (91,68)) corresponding to the position of the nose and the vector **305** (the value is (15,4)) are taken as an example for illustrating.

(36) In the step **S1001**, the first component 15 of the vector **305** is substituted by the second calculation module **103** into the coefficient $\alpha_{\text{sub.1}}$ of the first variable x of the above intermediate two-variable linear equation (1) and the second component 4 of the vector **305** is substituted into the coefficient $\beta_{\text{sub.1}}$ of the second variable y of the above intermediate two-variable linear equation (1) to obtain an equation $15x + 4y + \gamma_{\text{sub.1}} = 0$. In the step **S1002**, a first coordinate component 91 and a second coordinate component 68 of the third keypoint **304** are respectively substituted by the second calculation module **103** into the first variable x and the second variable y of the above intermediate two-variable linear equation respectively to obtain the constant term $\gamma_{\text{sub.1}}$ of the intermediate two-variable linear equation to be -1637 . In the step **1003**, the first component of the vector **305** is substituted into the coefficient $\alpha_{\text{sub.2}}$ of the first variable x of the above two-variable linear function (2), the second component of the vector **305** is substituted into the coefficient $\beta_{\text{sub.2}}$ of the second variable y of the above two-variable linear function (2), and the constant term $\gamma_{\text{sub.1}} = -1637$ of the above intermediate two-variable linear equation is substituted into the constant term $\gamma_{\text{sub.2}}$ of the above two-variable linear function (2) to obtain the two-variable linear function $15x + 4y - 1637$.

(37) It is to be noted that the coordinates (73,51) of the first keypoint **302** are respectively substituted by the determination module **104** into the first variable x and the second variable y of the two-variable linear function $15x + 4y - 1637$ obtain a first value of -338 , and the coordinates (88,55) of the second keypoint **303** are substituted into the two-variable linear function $15x + 4y - 1637$ to obtain a second value of -97 . The face in the image **301** is determined as the side face state by the determination module **104** responding to the product of the first value -338 and the second value -97 as a positive number.

(38) FIG. 6 is a structure schematic diagram of an electronic device drawn according to some embodiments of the present disclosure. As shown in FIG. 6, on a hardware level, the electronic device **600** includes processors **601-1**, **601-2** to **601-R**, an internal memory **602** and a non-volatile memory **603**, and R is a positive integer. The internal memory **602** is a random-access memory (RAM) for example. The non-volatile memory **603** is at least 1 disc memory and the like for example. Certainly, the electronic device **600** may further include hardware required by other functions.

(39) The internal memory **602** and the non-volatile memory **603** are configured to store programs, the programs can include program codes, and the program codes include computer operation instructions. The internal memory **602** and the non-volatile memory **603** provide instructions and data for the processors **601-1** to **601-R**. The processors **601-1** to **601-R** read corresponding

computer programs from the non-volatile memory **603** into the internal memory **602** and then run the programs, the detection system **100** is formed on a logic level, and the processors **601-1** to **601-R** are specifically configured to execute steps recorded in FIGS. 7-10. Certainly, the modules of the detection system **100** can also be made in a hardware mode, and the present disclosure does not limit it.

(40) The processors **601-1** to **601-R** may be integrated circuit chips and have signal processing capability. During implementation, the methods and steps disclosed in the above embodiments can be completed through hardware integrated logic circuits or software instructions in the processors **601-1** to **601-R**. The processors **601-1** to **601-R** can be universal processors and include a central processing unit (CPU), a tensor processing unit, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic apparatuses. The processors can implement or perform the methods and the steps disclosed in the foregoing embodiments.

(41) In some embodiments of the present disclosure, a computer readable storage medium is also provided. The computer readable storage medium stores at least one instruction. When the at least one instruction is executed by the processors **601-1** to **601-R** of the electronic device **600**, the processors **601-1** to **601-R** of the electronic device **600** can execute the methods and steps disclosed in the above embodiments.

(42) Examples of the computer storage medium include, but are not limited to, phase change RAM (PRAM), static random access memories (SRAM), dynamic random access memories (DRAM), other types of random access memories (RAM), read-only memories (ROM), electrically erasable programmable read-only memories (EEPROM), flash memories or other internal memory technologies, compact disc read-only memories (CD-ROM), digital versatile discs (DVD) or other optical memories, magnetic tape cassettes, magnetic tape disc memories or other magnetic storage devices or any other non-transmission media, and can be configured to store information which can be accessed by computing devices. According to the definition here, the computer readable medium does not include transitory media, such as modulated data signals and carriers.

(43) Based on the above detection system and the detection method provided by some embodiments of the present disclosure, the state of the face can be quickly determined by performing simple algebraic operation and determination on the first keypoint, the second keypoint and the third keypoint of the detected face.

(44) Although the present disclosure has been described in considerable detail with reference to certain preferred embodiments thereof, the disclosure is not for limiting the scope of the invention. Persons having ordinary skill in the art may make various modifications and changes without departing from the scope and spirit of the disclosure. Therefore, the scope of the appended claims should not be limited to the description of the preferred embodiments described above.

Claims

1. A detection system, comprising: a keypoint acquisition module, configured to receive an image containing a face and obtain a first keypoint, a second keypoint and a third keypoint of the face based on the image, the keypoint acquisition module obtaining the third keypoint based on a predetermined position on a midline of a human face and the first keypoint and the second keypoint based on two predetermined positions paired outside the midline of the human face; a first calculation module, configured to obtain a vector based on the first keypoint and the second keypoint; a second calculation module, configured to obtain a two-variable linear function based on the vector and the third keypoint; and a determination module, configured to substitute coordinates of the first keypoint and coordinates of the second keypoint into the two-variable linear function to obtain first and second values, respectively, and determine the state of the face based on the first and second values.

2. The detection system according to claim 1, wherein the determination module is configured to determine the face as a side face state by responding to the product of the first and second values as a positive number, and determine the face as a front face state by responding to the product of the first and second values as a negative number.
3. The detection system according to claim 1, wherein the first calculation module is configured to subtract the coordinates of the first keypoint from the coordinates of the second keypoint to obtain the vector.
4. The detection system according to claim 1, wherein the second calculation module executes the following steps to obtain the two-variable linear function: setting a first component of the vector as a coefficient of a first variable of an intermediate two-variable linear equation, and setting a second component of the vector as a coefficient of a second variable of the intermediate two-variable linear equation; substituting a first coordinate component and a second coordinate component of the third keypoint into the intermediate two-variable linear equation to obtain a constant term of the intermediate two-variable linear equation; and setting the first component of the vector as a coefficient of a first variable of the two-variable linear function, setting the second component of the vector as a coefficient of a second variable of the two-variable linear function, and setting the constant term of the intermediate two-variable linear equation as a constant term of the two-variable linear function.
5. The detection system according to claim 1, wherein the predetermined position on the midline of the human face is the position of the nose.
6. The detection system according to claim 1, wherein the two predetermined positions paired outside the midline of the human face are the positions of the eyes.
7. The detection system according to claim 1, wherein the keypoint acquisition module comprises a neural network module, and the neural network module is configured to receive the image of the face and output the coordinates of the first keypoint, the coordinates of the second keypoint and the coordinates of the third keypoint.
8. A detection method applicable to a detection system comprising a keypoint acquisition module, a first calculation module, a second calculation module and a determination module, comprising the following steps: (a) receiving an image containing a face by the keypoint acquisition module, and obtaining a first keypoint, a second keypoint and a third keypoint of the face based on the image, wherein the keypoint acquisition module obtains the third keypoint based on a predetermined position on a midline of a human face and the first keypoint and the second keypoint based on two predetermined positions paired outside the midline of the human face; (b) obtaining a vector by the first calculation module based on the first keypoint and the second keypoint; (c) obtaining a two-variable linear function by the second calculation module based on the vector and the third keypoint; and (d) substituting, by the determination module, the coordinates of the first keypoint and the coordinates of the second keypoint into the two-variable linear function to obtain first and second values, respectively, and determining the state of the face based on the first and second values.
9. The detection method according to claim 8, wherein the above step (d) comprises: (d1) determining the face as a side face state by the determination module responding to the product of the first and second values as a positive number; and determining the face as a front face state by the determination module responding to the product of the first and second values as a negative number.
10. The detection method according to claim 8, wherein the above step (b) comprises: (b1) subtracting, by the first calculation module, the coordinates of the first keypoint from the coordinates of the second keypoint to obtain the vector.
11. The detection method according to claim 8, wherein the above step (c) comprises the following steps executed by the second calculation module: (c1) setting a first component of the vector as a coefficient of a first variable of an intermediate two-variable linear equation, and setting a second

component of the vector as a coefficient of a second variable of the intermediate two-variable linear equation; (c2) substituting a first coordinate component and a second coordinate component of the third keypoint into the intermediate two-variable linear equation to obtain a constant term of the intermediate two-variable linear equation; and (c3) setting the first component of the vector as a coefficient of a first variable of the two-variable linear function, setting the second component of the vector as a coefficient of a second variable of the two-variable linear function, and setting the constant term of the intermediate two-variable linear equation as a constant term of the two-variable linear function.

12. The detection method according to claim 8, wherein the predetermined position on the midline of the human face is the position of the nose.

13. The detection method according to claim 8, wherein the two predetermined positions paired outside the midline of the human face are the positions of the eyes.

14. The detection method according to claim 8, wherein the keypoint acquisition module comprises a neural network module, the neural network module being configured to receive the image of the face and output the coordinates of the first keypoint, the coordinates of the second keypoint and the coordinates of the third keypoint.
